PROJECT MANUAL/SPECIFICATIONS

A NEW FACILITY FOR
MAGNOLIA ELECTRIC POWER ASSOCIATION

McCOMB, MS 39648

FISHER ARNOLD

ENGINEERING INTEGRATION

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SECTION 05 10 00 - STRUCTURAL STEEL

PART 1 - GENERAL

1.01 SCOPE OF WORK

A. Furnishing of and paying for all labor, materials, services, appliances, and equipment necessary for the execution, installation, and completion of all work specified herein.

B. Section included:
   1. Fabrication and erection of all structural steel as shown on the drawings and/or specified herein, including but not necessarily limited to the following:
      a. All structural steel rolled sections such as beams, columns, channels, angles, anchor plates, bearing plates, brackets, braces, anchor bolts, sleeves, inserts, and/or other items incidental to construction as shown on the Structural (S) Drawings or as normally required to complete the erection and support of all structural steel work specified herein.
   2. Anchor bolts, sleeves, bearing plates, and other incidental metal items of structural and/or miscellaneous steel as and where shown on the drawings and shall be provided as indicated or specified. At the proper time these shall be furnished to the respective contractor, where applicable, including instructions or templates for their installation within concrete construction.
   3. All miscellaneous steel members supporting penetrations of roof deck whether or not same is shown on the drawings.
   4. All connections, erection fittings, and devices required to complete this work.
   5. Shop painting and field touch-up painting as noted on drawings and in part 2.01 of this specification.
   6. All miscellaneous steel angles, beams, rods, and/or braces to be furnished and/or installed for support or bracing of door frames, operable walls, and/or any other members as and where shown on the drawings.
   7. All other miscellaneous steel appearance items complete with brackets, etc., for complete installation.

C. Work finished but installed under other sections:
   1. Anchorages cast in concrete: Section 033000

1.02 RELATED SECTIONS

A. Section 013400 - Submittals: Shop Drawings, Product Data, and Samples.

B. Section 014100 - Testing Laboratory Services

C. Section 033000 - Cast-In-Place Concrete: Including Grouting of Baseplates and Bearing Plates

D. Section 053150 - Composite Steel Deck: Including Support Framing for Small Openings up to 12 inches in size.

E. Section 055000: Metal Fabrications

F. Section 099000: Finish Painting

G. Section 131000: Pre-Engineered Steel Building

1.03 RELATED DOCUMENTS

A. Drawings and general provisions of contract, including General and Supplementary Conditions and Division 1 - "General Requirements" apply to work of this section.

B. Use this specification in conjunction with the General Notes and specific directives on the Contract Structural Drawings.

C. Perform work in accordance with the AISC Specifications and Code of Standard Practice except as modified herein.

1.04 REFERENCES

A. International Building Code (2006).

B. American Institute of Steel Construction (AISC):
   1. "Manual of Steel Construction" and included "Specifications for the Design, Fabrication, and Erection of Structural Steel for Buildings" including "Commentary" and supplements thereto as issued and AISC Code of Standard Practice

C. American Welding Society (AWS):
   1. "Structural Welding Code" (AWS D1.1) including the "Standard Qualification Procedure", Section 5. qualifications, Part B, Procedure Qualification.
   2. "Guide for the Non-destructive Inspection of Welds" (ANSI/AWS B1.10).
   3. Guide to "Standard for Qualification and Certification of Welding Inspectors" (AWS-QC1).
   4. Welding Inspection (AWS 2nd Edition, WI).

D. Research Council on Structural Connections of the Engineering Foundation:
   1. Specifications for Structural Joints Using ASTM A325 or A490 Bolts.

E. American Society for Testing and Materials (ASTM):
   1. ASTM A6 "General Regulations for Delivery of Rolled Steel Plates, Shapes, Sheet Piling, and Bars for Structural Use"
   2. See Materials Specifications listed under Part 2 - PRODUCTS

F. Steel Structures Painting Council (SSPC): "Steel Structures Painting Manual": Vols. 1 and Vol. 2 ; Systems and Specifications.

G. Industrial Fasteners Institute: Handbook for Bolt, Nut, and Rivet Standards.

H. American Society for Non-Destructive Testing (ASNT), Recommended Practice SNT-TC-1A.

I. American Council of Independent Laboratories Inc. (ACIL) "Manual of Practice".
   1. Quality Control System, Requirements for a Testing and Inspection Laboratory.
   2. Scope of Services, Materials, Engineering Testing and Inspections.

J. Alterations of Contractual Relationship: No provision of any referenced standard specification, manual or code (whether or not specifically incorporated by reference in the Contract Documents) shall be effective to change the duties and responsibilities of OWNER, CONTRACTOR, or ENGINEER, or any of their consultants, agents or employees from those set forth in the Contract Documents, nor shall it be effective to assign to ENGINEER, or any other authority to supervise or direct the furnishing or performance of the work or any duty or authority to undertake responsibility contrary to the provisions of these Contract Documents.

1.05 SUBMITTALS

A. Submit under provisions of Section 01300.

B. Shop Drawings: Submit Shop and Erection Drawings, as specified in AISC Code and AISC Specifications clearly indicating profiles, sizes, spacing, and locations of structural members, connections, attachments, anchorages, framed openings larger than 12" square, size and type of fasteners, cambers, shop painting, and all other details required for the fabrication and erection of the structural steel. Provide setting drawings, templates, and directions for installation of anchor bolts and other anchorages to be furnished and installed for work in other sections.

C. Welding Procedure: Indicate welded connections with AWS A2.1 welding symbols; indicate net weld size and lengths. Submit written description required to illustrate each welding procedure to be performed. Submit descriptive data for field welding equipment including type, voltage, and amperage.

D. Erection Procedure: Submit descriptive data to illustrate the structural steel erection procedure including the sequence of erection and temporary staying and bracing.

E. Mill Test Reports: Submit under provisions of Specifications Section 01400, Manufacturer's Certificates, indicating structural strength, destructive and non-destructive test analysis.

F. Manufacturer's Mill Certificate: Submit under provisions of Specification Section 01400 certifying that the materials meet or exceed specified requirements.

G. Welder's Certificates: Submit certificates certifying welders employed on the work meet AWS qualifications within the previous 12 months.

H. Surveys: Submit certified copies of each survey conducted by a registered P.E., showing elevations and locations of base plates and/or anchor bolts to receive structural steel, and final elevations and locations for major members. Indicate discrepancies between installation and contract documents.

I. Proofs of compliance for materials: Submit for approval the following:
   1. Reports of ladle analysis for all steels.
   2. Reports of tensile properties and bend tests for steel shapes, bars, and plates.
   3. Certified Mill Test Reports required by applicable ASTM Material Specifications (AISC Code, Div. 5).
   4. Submit Certificates of Conformance for:
      a. Structural steel tubing.
      b. Shear studs in accordance with AWS D1.1.
      c. Filler material for welding.
   5. Reports of mechanical properties of headed stud type shear connectors.
   6. Reports of mechanical tests for high-strength threaded fasteners.

J. Manufacturer's literature: Submit description of each type of welding stud and arc shield.

K. Inspection reports: Submit reports for the inspection and tests specified in 1.08 below.

L. Certification of Quality Assurance Program (See 1.08)

1.06 PROJECT RECORD DOCUMENTS

A. Comply with the provision of Section 01720.

B. Submittals

C. Certifications

1.07 RESPONSIBILITIES

A. The A/E is responsible for the design and the preparation of the Contract Documents. The Contractor/Fabricator is responsible for properly implementing the design drawings, properly furnishing materials and workmanship, maintaining the specified fabrications and erection tolerances, and for fit and erectability of the structure. The A/E is not responsible for reviewing these aspects, nor for reviewing the means and methods of construction or related job safety issues, which are the Contractor's responsibility.

B. AISC Code, paragraph 1.5.1. Where connections are not shown on drawings, the connections shall be in accordance with the requirements of the AISC Specifications Sections 1.15 Connections. Where reactions and/or moments are shown on drawings, connections shall be designed by the fabricator to accommodate these, subject to review and approval by the Engineer.

C. Substitutions of member sizes due to non-availability of materials shall be of equivalent strength and rigidity to that specified, shall be compatible with the design and shall be approved by the Engineer after being specifically called to his attention in writing.

1.08 QUALITY ASSURANCE

A. General Requirements: The general requirements for Quality Control and Assurance are outlined in the following AISC documents:
   1. "Code of Standard Practice for Steel Buildings and Bridges", Section 8 - Quality Assurance.
   2. "Specifications for Design, Fabrication, and Erection of Structural Steel for Buildings", including "Commentary" and supplements there to as issued.
   3. "Specifications for Architecturally Exposed Structural Steel", AISC.

B. Qualifications: The structural steel fabricating plant and the steel erector shall have the personnel, organization, experience, procedures, knowledge, equipment, capacity, and commitment to produce and erect fabricated steel of the structural steel work involved in this project. Certification under the AISC "Quality Certification Program", SBCCI report No. Q.A. 7801 or equivalent verification for the following categories will suffice: I - Conventional Steel Structures.

C. Quality Control Criteria: The Quality Criteria and Inspection Standards for the preparation of materials, fitting and fastening, dimensional tolerances, surface preparation and painting, non-destructive examination and special fabrication problems shall conform to the AISC recommendations contained in Publication No. S323 (2P4M1181) "Quality Criteria and Inspection Standards". Mill material camber, flatness, straightness, sweep, and cross section are established by ASTM A6. The fabricator is allowed to use controlled heating, mechanical straightening, or a combination of both methods, consistent with the manufacturer's recommendations, to adjust for conformance.

D. Quality Assurance: Both the Fabricator and Erector shall maintain a Quality Assurance program conforming to AISC Code of Standard Practice, Section 8, and AISC Specification for the Design, Fabrication and Erection of Structure Steel for Buildings, Section 1.26, to assure that all of the work is performed in accordance with this contract document. It shall be the responsibility of the Contractor to maintain control of the quality of the materials and workmanship, and conformance to the project Specifications. Materials and fabrication procedures are subject to inspections and tests in mill, shop, and field, conducted by a qualified inspection agency. Such inspections and tests will not relieve Contractor of responsibility for providing materials and fabrication procedures in compliance with specified requirements.

E. Material Certification: The Fabricator shall provide an affidavit stating that the structural steel furnished meets the requirements of this specification. Certified mill test reports or certified reports of tests made by the fabricator or a testing laboratory in accordance with

ASTM A6 or A568, as applicable, shall constitute sufficient evidence of conformity with the above standards of the identity of these materials are properly maintained.

F. Welding (Shop and Field): Fabrication/erection inspection and testing of weldments shall be provided by the Contractor in accordance with AWS D1.1. The fabrication/ erection inspector(s) shall be AWS Certified Welding Inspector(s) in accordance with the provisions of AWS QCI, Standard for Qualifications and Certification of Welding Inspectors. Provide certification that welding inspectors, welders, welding operators, and tack welders to be employed in work have satisfactorily passed AWS qualification tests within previous twelve (12) months prior to starting the work. If recertification of welders is required, it will be Contractor's responsibility to assure compliance. Each welder working on the project shall mark his identification symbol at each weldment completed whether in shop or field.

G. High-Strength Bolts: Shall be installed as specified in ASTM A325 and A490. Each member of bolting crew applying high - strength bolts shall be assigned an identifications mark or symbol which he shall apply to each joint worked. The Contractor shall verify that all contact surfaces in friction-type connections conform to the applicable requirements, and that all bolts have been appropriately installed and tightened.

H. The special inspector shall perform the following verification and inspection during steel construction:
   1. Verify material of high-strength bolts, nuts and washers by identification markings to conform to ASTM standards specified in the construction documents and manufacturer’s certificate of compliance.
   2. Verify material of structural by identification markings to conform to ASTM standards specified in the construction documents and manufacturer’s certificate of compliance.
   3. Verify framing layout, size and details according to construction documents and approved shop drawings.
   4. While the work is in progress, verify that the requirements for bolts, nuts, washers, and paint; bolted parts; and installation and tightening are met as indicated in construction documents and standards.
   5. Verify weld filler material by identifying markings to conform to AWS specification in the approved construction documents and manufacturer’s certificate of compliance.
   6. Inspect welding of structural steel as follows:
      a. Continuously inspect complete and partial penetration groove welds, multi-pass fillet welds and single pass fillet welds greater than 5/16".
      b. Periodically inspect single pass fillet welds less than or equal to 5/16" and floor and deck welds.
   7. Inspect steel frame and bracing joint details for compliance with approved construction documents, such as bracing, stiffening, member locations, gussets and application of joint details at each connection.

I. The special inspector shall verify that the fabricator maintains detailed fabrication and quality control procedures that provide a basis for inspection control of the workmanship and the fabricator’s ability to conform to approved construction documents and referenced standards. The special inspector shall review the procedures for completeness and adequacy relative to the code requirements for the fabricator’s scope of work.

1.09 QUALIFICATIONS

A. Fabricator and Erector shall have completed a project of similar scope and shall have adequate facilities, personnel, and equipment to meet production and quality requirements and to maintain proper job progress. Certification by the AISC Quality Certification Program will provide satisfactory evidence of compliance.

B. Welding procedures, welders, welding operations, and tackers shall be qualified in accordance with the AWS Structural Welding Code, Section 5.

C. Periodic Structural Observation: (Spec. Section 01400). The design engineer of record or his designated representative will make visits to the site during construction to observe general compliance with the approved plans, specifications, and change orders, upon request by the Owner.

D. The Contractor shall correct deficiencies in structural steel work which inspections and laboratory test reports have indicated to be not in compliance with requirements. Perform additional tests, at Contractor's expense, as may be necessary to reconfirm any non-compliance of original work and as may be necessary to show compliance of corrected work.

1.10 DELIVERY, STORAGE AND HANDLING:

A. Deliver, store, protect and handle products to site under provisions of Section 01600.

B. Delivery of materials to be installed under other sections:
1. Anchor bolts and other anchorage devices which are embedded in cast-in-place concrete or masonry construction shall be delivered to the project site in time to be installed before the start of cast-in-place concrete operations or masonry work.
2. Provide setting drawings, templates, and directions for the installation of the anchor bolts and other devices.

C. Storage of materials:
1. Structural steel members which are stored at the project site shall be above ground on platforms, skids, or other supports.
2. Steel shall be protected from corrosion.
3. Other materials shall be stored in a weathertight and dry place until ready for use in the work.
4. Packaged materials shall be stored in their original, unbroken packages or containers.

PART 2 - PRODUCTS

2.01 MATERIALS AND COMPONENTS

A. Refer to AISC Specifications, Section 1.4 - Materials: See Structural Drawings and General Notes for specific requirements, unless modified or specified hereafter.

B. Structural Steel: ASTM A36 or ASTM A572 - Grade 50 or as specified on drawings. See AISC Specifications, Section 1.4.1 Structural Steel for other grades of steel shown on drawings.

C. Bolts: AISC Specifications, Section 1.4.4 - Bolts
1. Unfinished Bolts: ASTM A307
2. High-Strength Steel Bolts: ASTM A 325, A490, or A449.
3. Anchor Bolts and Nuts: ASTM A307, Grade A .
4. High - strength Anchor Bolts: ASTM A490.
5. All faying surfaces shall be prepared as required for Class A or better slip-critical joints in connections that are part of the Seismic Force Resisting System.

D. Filler metal and flux for welding: AISC Specifications, Section 1.4.5.
1. AWS A5.1, E70 unless otherwise approved. Electrodes shall be compatible with the base material being welded. Manufacturer's certification evidence of conformance.
2. All welds used in Seismic Force Resisting System shall be made with a filler metal that has a minimum CVN toughness of 20 ft-lbs at minus 20 degrees F.

E. Non-shrink grout for columns and other bearing plates: Section 03300 - Concrete.

F. Shop paint primer: AISC Code of Standard Practice, AISC Specifications and AISC-SSPC "Systems and Specs."
   1. Except for contact surfaces, surfaces inaccessible after shop assembly shall be cleaned and painted prior to assembly.
   2. Surfaces within 2 inches of any weld location shall be free of materials that would prevent proper welding or produce toxic fumes during welding.
   3. Paint is permitted in bearing type connections.
   4. Paint on steel members which are to receive fireproofing shall be compatible with spray on fireproofing.

G. Other materials shall conform to the applicable current specifications of the ASTM.

2.02 FABRICATION

A. Conform to AISC Code of Standard Practice, Section 6 and AISC Specifications, Section 1.23.

B. Shop fabrication and assembly: Fabricate and assemble structural assemblies in shop to greatest extent possible. Fabricate items of structural steel in accordance with approved shop drawings. Properly mark and match-mark materials for field assembly. Fabricate for delivery sequence which will expedite erection and minimize field handling of materials. Verify all dimensions prior to fabrication.

C. Dimensional and straightness tolerances: Conform to AISC Code, Section 6.4 and ASTM A6 for compression joints requiring milled ends.

D. Provide openings in structural members for other building components as shown on drawings. Locate holes so as not to cause any appreciable reduction in strength of members and reinforce openings with steel plates and/or angles sized and welded in place to restore members to original strength as approved by the Engineer.

E. All openings through steel roof deck are to receive supplemental steel framing. See plans.

F. Cut and mill column ends and bearing plates accurately to assure full contact of bearing surfaces prior to welding.

G. Camber: Horizontal members to accommodate dead-plus live load deflections as indicated on the contract drawings. See General Notes on Structural Drawings, AISC Specifications and AISC Code. Camber for composite beams and girders. See structural plans.

H. Clean, prepare, and shop prime structural steel members. Do not prime surfaces to be field-welded or to be bolted or in contact with concrete.

I. Shop and field connections shall be riveted, bolted, welded, or a combination of these as required to conform to AISC Specifications.
   Field connections:
   1. Bolted in accord with AISC Specifications except where welded connections are required or selected.
   2. High-strength threaded fasteners shall be used for bolted connections except where standard threaded fasteners are permitted.
   3. Welded connections shall conform to AWS D1.1. Each complete and partial penetration weld shall be tested for the full length of the weld by ultrasonic or other approved non-destructive methods. Welding materials shall be of the type required for materials being welded and conforming to applicable AWS Specifications.

J. Bearing plates:
   1. Bearing plates shall be provided under beams, closures, and girders resting on footings, piers, and walls as shown on the drawings with anchorage devices.
   2. Bearing plates shall be either attached or loose as required for erection.

K. Metal framing appearance items:
   1. Form members to Manufacturer's standard shapes meeting design criteria.
   2. Cut right angle connections of framing components to fit squarely against abutting members as shown on drawings. Weld or bolt as shown.
   3. Grind smooth all welds and holes cut in structural members exposed to view after fabricating.
   4. All items are to be square, straight, and true to detail in all respects.
   5. Fabricate all items as shown on drawings and specified herein. All must present smooth surface for use and appearance.

L. All welded joints that are part of the Seismic Force Resisting System (such as moment resisting beam-column connections, bracing connections, column base plate connections of columns that are part of moment frame or braced frames, knee braces, etc.) shall meet the following requirements.

   1. Welding shall be performed in accordance with a Welding Procedure Specification (WPS) as required in AWS D1.1. Contractor shall submit a written WPS to A/E for approval prior to start of fabrication.

   2. All welds shall be made with a filler metal that has a minimum CVN toughness of 20 ft-lbs at minus 20 degrees F, as determined by AWS classification or manufacturer certification.

   3. Discontinuities created in the weld by errors or by fabrication or erection operations, such as tack welds, erection aids, air-arc gouging, and flame cutting, shall be repaired. The contractor shall submit a repair procedure to the A/E for approval.

PART 3 - EXECUTION

3.01 INSPECTION

A. Erector shall examine areas and conditions under which structural steel work is to be installed and notify Contractor in writing of conditions detrimental to proper and timely completion of work.

B. Do not proceed with work until unsatisfactory conditions have been corrected in an acceptable manner.

3.02 ERECTION

A. Erector shall check elevations of concrete and masonry bearing surfaces and locations of anchor bolts and similar devices before erection work proceeds and report discrepancies to Contractor and Engineer.

B. Do not proceed with erection until corrections have been made or until compensating adjustments to structural steel work have been agreed upon with Architect.

C. Erect structural steel in accordance with the drawings and as provided in AISC Code and AISC Specs., except as hereinafter specified.

D. Field assembly:

1. Steel frames and/or members shall be assembled accurately to the lines and elevations indicated and within the erection tolerances specified in AISC Code, Section 7.11.
2. The various members forming parts of a complete frame or structure after being assembled shall be aligned and adjusted accurately before being fastened.
3. Fastening of splices of compression members shall be done after the abutting surfaces have been brought completely into contact.
4. Bearing surfaces and surfaces which will be in permanent contact shall be cleaned before the members are assembled.
5. Splices shall be permitted only where indicated or approved in writing.

E. Bearing plates:
1. Bearing plates for columns, beams, and similar structural members shall be aligned with wedges or shims before grouting.

F. Make adequate provisions for all erection loads and for sufficient temporary bracing to maintain structure safe, plumb, and in true alignment until completion of erection and installation of necessary permanent bracing.

G. Do not field-cut or alter structural members without the written approval of the Engineer.

H. Temporary shoring and bracing: Provide temporary shoring and bracing members with connections of sufficient strength to bear imposed loads. Remove temporary members and connections when permanent members are in place and final connections are made. Provide temporary guy lines to achieve proper alignment of structures as erection proceeds.

I. Temporary planking: Provide temporary planking and working platforms as necessary to complete work effectively.

J. Anchor bolts: Furnish anchor bolts and other connectors required for securing structural steel to foundations and other in-place work. Furnish templates and other devices as necessary for presetting bolts and other anchors to accurate locations.

K. Setting baseplates and bearing plates: Clean concrete and masonry bearing surfaces of bond-reducing materials and roughen to improve bond to surfaces. Clean bottom surface or baseplates and bearing plates. Set loose and attached baseplates and bearing plates for structural members on wedges or other adjusting devices.

L. Tighten anchor bolts after supported members have been positioned and plumbed. Do not remove wedges or shims, but if protruding, cut off flush with edge of baseplate or bearing plate prior to packing with grout.

M. Splice members only where indicated and accepted on approved shop drawings unless specifically approved in writing.

N. Erection Bolts: On exposed, welded construction, remove erection bolts, fill holes with plug welds, and grind smooth at exposed surfaces.

O. Comply with AISC Specifications for bearing, adequacy of temporary connections, alignment, and removal of paint on surfaces adjacent to field welds. Do not enlarge holes in members by burning or by use of drift pins except in secondary bracing members. Ream holes which must be enlarged to admit bolts.

P. Gas cutting: Do not use gas cutting torches in field for correcting fabrication errors in structural framing. Cutting will be permitted only on secondary members which are not under stress, as acceptable to Engineer. Finish gas-cut sections equal to a sheared appearance when permitted.

3.03 PROTECTION

A. Do not use structural units for storage or working platforms until permanently secured in position.

B. Assure that construction loads do not exceed carrying capacity of structure.

3.04 FIELD PAINTING TOUCH-UP

A. After the erection of all structural and miscellaneous steel members, touch-up paint all abrasions and unpainted areas with the same paint used for the shop painting.

B. All shelf angles and lintels which will have any part exposed to weather after completion of work will receive prior to erection one finish paint coat by Section 09900 - Painting.

3.05 CLEANING

A. Remove from the work site all debris caused by this sub-contract.

END OF SECTION